BUILD WITH AI:
INTELLIGENT STRUCTURAL ANALYSIS FOR MODERN CONSTRUCTION
(Civil Engineering Insight Studio)

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1. INTRODUCTION

1.1 Project Overview

Civil Engineering Insight Studio is an AI-powered web application designed to analyze images of civil engineering structures and generate detailed structural descriptions automatically. The system takes user input in the form of a construction image and optional prompt, and processes it using a generative AI vision model to produce structured engineering insights.

The application uses Streamlit for the frontend to provide a simple and interactive user interface, while Python handles backend processing. By integrating Google Gemini Pro Vision, the project demonstrates how modern AI technologies can be applied to automate structural documentation and analysis in civil engineering.

1.2 Purpose

The primary purpose of this project is to reduce the time and effort involved in manually describing construction structures. It aims to provide engineers and construction professionals with automated material identification, structural component detection, and project progress documentation.

Additionally, the project serves as a practical implementation of generative AI in the civil engineering domain, showcasing how AI can enhance efficiency, accuracy, and reporting processes.

2. IDEATION PHASE

2.1 Problem Statement

Civil engineers often manually inspect images of structures and prepare reports describing materials, components, and construction stages. This process is time-consuming, subjective, and dependent on human expertise.

There is a need for an AI-based system that can automatically analyze structural images and generate professional engineering descriptions quickly and accurately.

2.2 Empathy Map Canvas

User: Civil Engineer / Construction Supervisor

- Thinks: I need faster and accurate structural documentation.
- Feels: Pressured due to deadlines and repetitive reporting work
- Needs: Automated analysis and reliable insights.

User: Project Manager

- Needs efficient documentation for project updates.
- Wants consistent and structured reports.

2.3 Brainstorming

Possible solutions included manual templates, predefined reporting formats, and rule-based image analysis systems. However, these methods lacked flexibility and intelligence.

The final solution selected was an AI-powered structural analysis system using Google Gemini Pro Vision to dynamically analyze images and generate detailed engineering descriptions.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User opens Streamlit app
2. Uploads civil engineering structure image
3. Enters optional prompt
4. Clicks “Describe Structure”
5. Backend sends data to Gemini Pro Vision
6. AI generates structural analysis
7. Output displayed to user

3.2 Solution Requirement

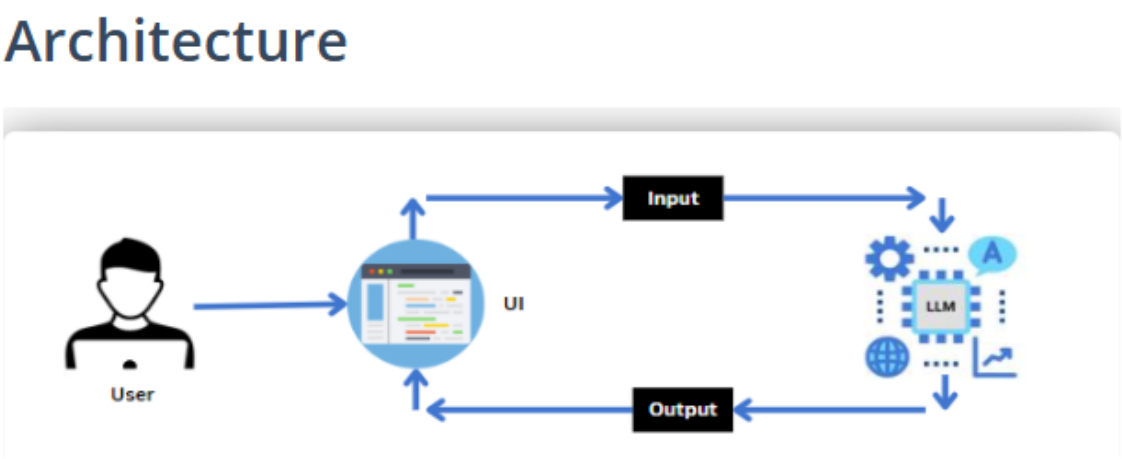
Functional Requirements:

- Accept image input (jpg, jpeg, png)
- Process image using AI model
- Identify materials (concrete, steel, bricks)
- Detect structural components (beams, columns, trusses)
- Generate structured engineering description
- Display output clearly

Non-Functional Requirements:

- Fast response time
- User-friendly interface
- Secure API integration

3.3 Data Flow Diagram



3.4 Technology Stack

The technology stack was selected to ensure simplicity, scalability, and effective AI integration.

Frontend: Streamlit

- **Backend**: Python

- **AI Model**: Google Gemini Pro V

- **Image Processing**: Pillow

- **Environment Management**: python-dotenv

- **Deployment**: Streamlit Cloud

4. PROJECT DESIGN

4.1 Problem-Solution Fit

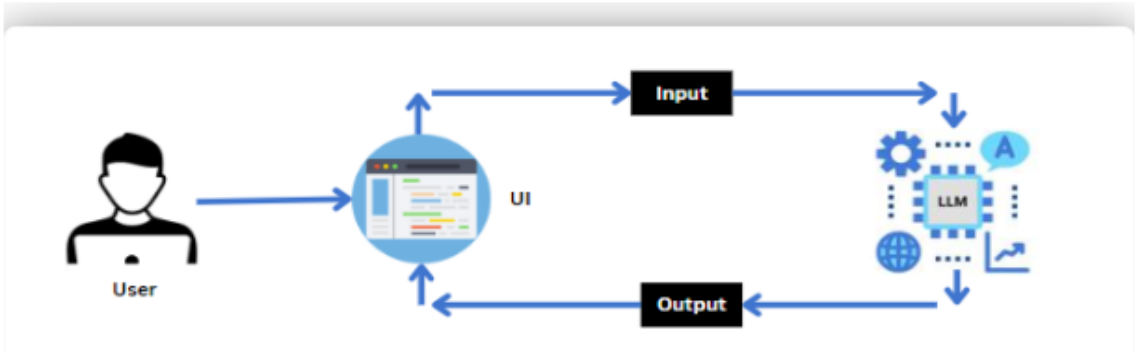
Manual structural documentation consumes time and may lead to inconsistencies. Civil Engineering Insight Studio provides a strong problem-solution fit by automating structural analysis using AI.This approach saves time, improves consistency, and enhances reporting efficiency in civil engineering projects.

4.2 Proposed Solution

The proposed solution is an AI-driven system that accepts images of civil engineering structures and generates comprehensive descriptions including:

- Structure type, Structural components
- Materials used, Construction methods
- Engineering challenges, Project stage

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

Phases:

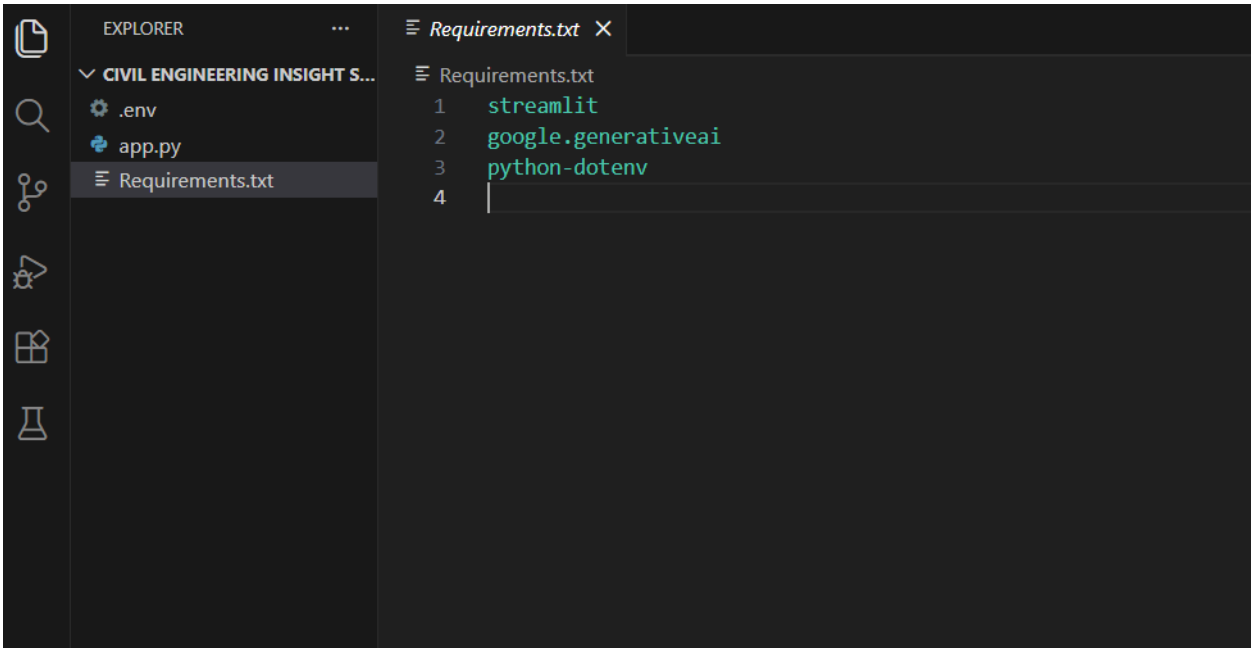
Research → Design → Development →API Integration → Testing → Deployment

6. FUNCTIONAL AND PERFORMANCE TESTING

The application was tested using multiple types of construction images including buildings, bridges, and site structures. Functional testing verified correct image upload handling and structured output generation. Performance testing ensured acceptable API response times and proper error handling when no image was uploaded. The testing phase confirmed that the system generates meaningful engineering insights under different scenarios.

7. RESULTS

The application successfully generates detailed structural analysis reports based on uploaded images. The system identifies materials, structural components, and construction stages, providing professional-level descriptions automatically.



EXPLORER

CIVIL ENGINEERING INSIGHT S...

- .env
- app.py
- Requirements.txt

app.py > ...

```python
from dotenv import load_dotenv
import streamlit as st
import os
import google.generativeai as genai
from PIL import Image

#load Api Key
load_dotenv()
genai.configure(api_key=os.getenv("GOOGLE_API_KEY"))

#Function to get gemini response
def get_gemini_response(input_text,image,prompt):
    model = genai.GenerativeModel('gemini-2.5-flash')
    response = model.generate_content([input_text,image[0],prompt])
    return response.text

def input_image_setup(uploaded_file):
    if uploaded_file is not None:
        bytes_data=uploaded_file.getvalue()
        image_parts = [
          {
              "mime_type": uploaded_file.type,
              "data":bytes_data
          }
        ]
        return image_parts
    else:
        raise FileNotFoundError("No file uploaded")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Rimcy\OneDrive\Desktop\Civil Engineering insight Studio>
PS C:\Users\Rimcy\OneDrive\Desktop\Civil Engineering insight Studio>
```

EXPLORER

CIVIL ENGINEERING INSIGHT S...

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```python
        raise FileNotFoundError("No file uploaded")

input_prompt="""
You are a civil engineering . Please describe the structure in the image and provide details such as its type,

1. Type of a structure - Description
2. Materials used - Description"""

#Intialize Streamlit app
st.set_page_config(page_title="Civil Engineering Insight Studio",page_icon="🏗️")
st.header("🏗️ Civil Engineering Insight Studio")
input_text=st.text_input("📝 Input Prompt:",key="input")
uploaded_file = st.file_uploader("🖼️ Choose an image...",type=["jpg","jpeg","png"])

if uploaded_file is not None:
    image =Image.open(uploaded_file)
    st.image(image,caption="Uploaded Image.",use_container_width=True)
submit = st.button("🚀 Describe Structure")

#If submit button is clicked
if submit:
    try:
        image_data=input_image_setup(uploaded_file)
        response = get_gemini_response(input_text,image_data,input_prompt)
        st.subheader("📋 Description of the Civil Engineering Structure:")
        st.markdown(f'<div class="st-ba">{response}</div>',unsafe_allow_html=True)
    except Exception as e:
        st.error(f"Error: {str(e)}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\Rimcy\OneDrive\Desktop\Civil Engineering insight Studio>
```

EXPLORER

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app.py > ...

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    if uploaded_file is not None:
        bytes_data=uploaded_file.getvalue()
        image_parts = [
          {
              "mime_type": uploaded_file.type,
              "data":bytes_data
          }
        ]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Rimcy\OneDrive\Desktop\Civil Engineering insight Studio> python -m streamlit run app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8502
  Network URL: http://192.168.1.3:8502
```

Civil Engineering Insight Studio

localhost:8502

Civil Engineering Insight Studio

Input Prompt:

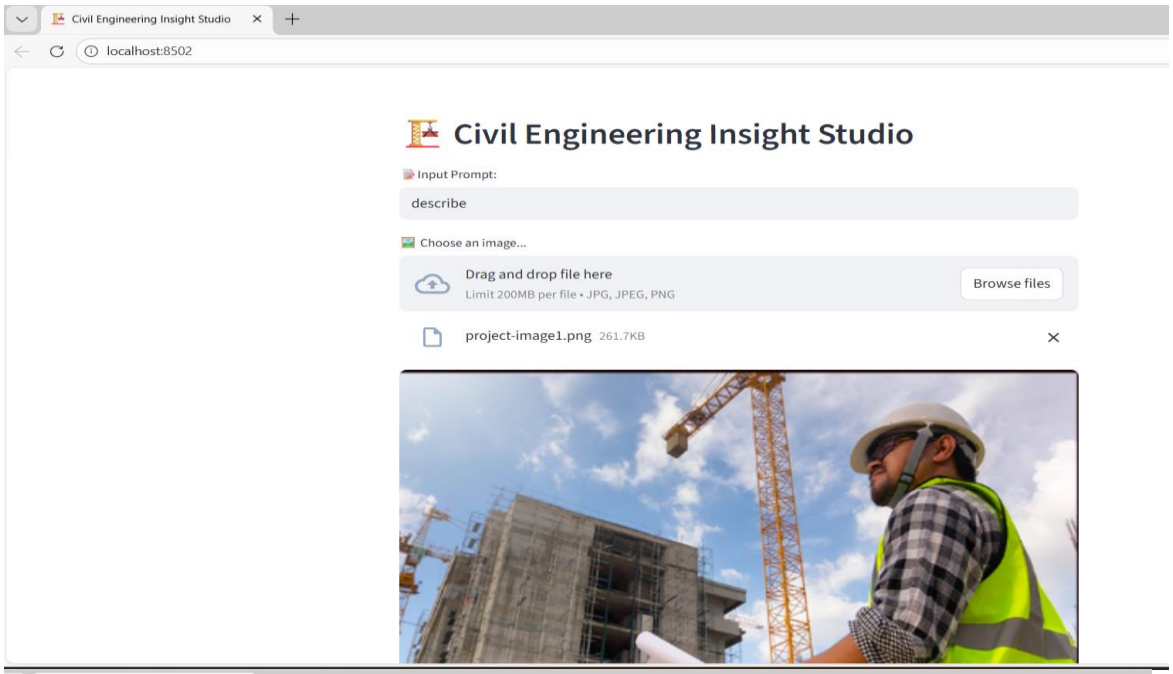
Choose an image...

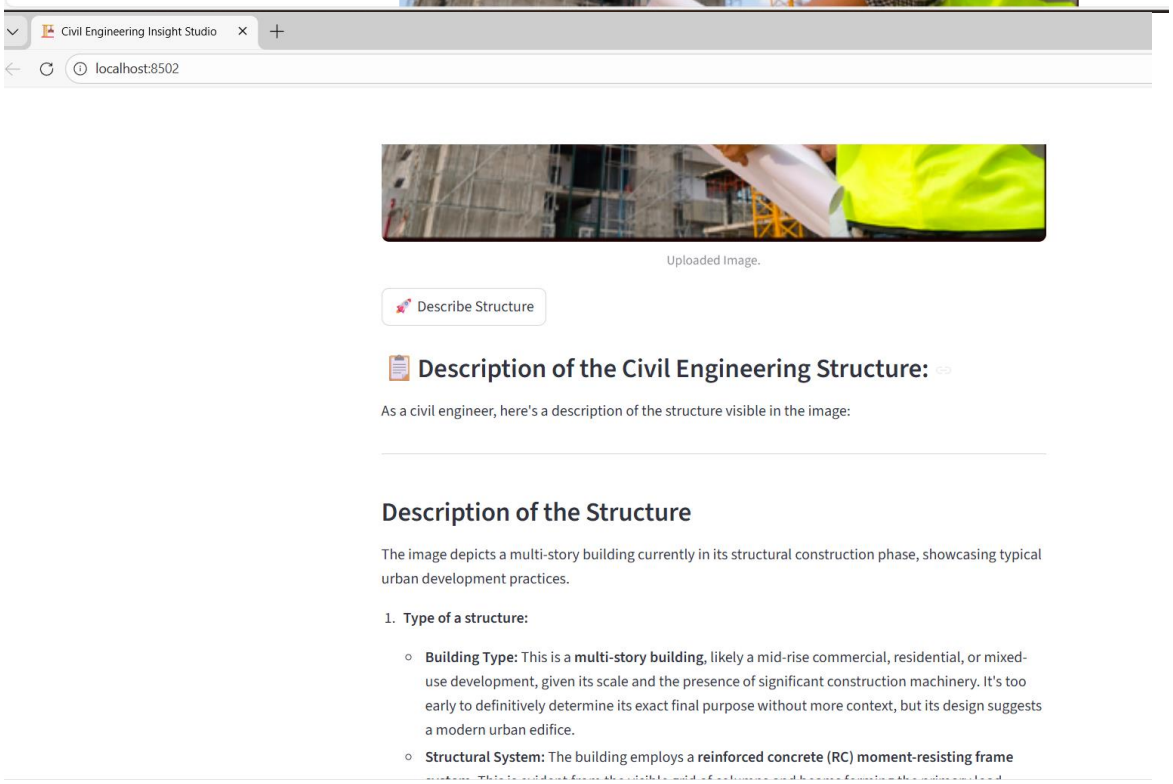
Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files

Describe Structure


Civil Engineering Insight Studio
localhost:8502
Civil Engineering Insight Studio
Input Prompt:
describe
Choose an image...
Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Browse files
project-image1.png 261.7KB


Civil Engineering Insight Studio
localhost:8502
Uploaded Image.
Describe Structure
Description of the Civil Engineering Structure:
As a civil engineer, here's a description of the structure visible in the image:
Description of the Structure
The image depicts a multi-story building currently in its structural construction phase, showcasing typical urban development practices.
1. Type of a structure:
Building Type: This is a multi-story building, likely a mid-rise commercial, residential, or mixed-use development, given its scale and the presence of significant construction machinery. It's too early to definitively determine its exact final purpose without more context, but its design suggests a modern urban edifice.
Structural System: The building employs a reinforced concrete (RC) moment-resisting frame

8. ADVANTAGES & DISA

DVANTAGES

Advantages:

- Saves documentation time
- AI-powered material detection
- Easy to use

Disadvantages:

- Requires internet connection
- Depends on API availability
- Accuracy depends on image quality

9. CONCLUSION

The Civil Engineering Insight Studio project demonstrates a practical application of generative AI in civil engineering. By integrating AI with a simple web interface, the system automates structural documentation and enhances efficiency.

The project highlights how AI technologies can be effectively applied in infrastructure and construction domains to improve productivity and decision-making.

10. FUTURE SCOPE

Future enhancements may include:

- Integration with BIM software
- 3D structural analysis
- Multi-image comparison
- Mobile application development

11. APPENDIX

GitHub Link:

https://github.com/BollaRimcy/BUILD-WITH-AI-INTELLIGENT-STRUCTURAL-ANALYSIS-FOR-MODERN-CONSTRUCTION

Demo Link:

https://drive.google.com/file/d/1SWkOABamOfczNkFoPMYlcixT9EvwtjJW/view?usp=drivesdk